

Wireless Charging System based on CC-CV Switching for Underwater Autonomous Vehicles

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Introduction

Autonomous underwater vehicles (AUVs), or underwater drones, are crucial for ocean exploration and infrastructure inspection. However, their limited battery capacity requires frequent manual recharging, posing risks and operational challenges. Moreover the current proposed methods tend to be a risk for the battery health of the AUVs. To address this, we propose developing a wireless power transfer (WPT) system for AUVs with CC-CV Mode Switching, enabling safer, contactless underwater charging and improving their operational efficiency.

Numerous studies have explored WPT for AUVs, focusing on coil design, power transfer efficiency, and impact of seawater as a transmission medium. Some key contributions include:

- **Gopika et al. (2023)**: Explored PSFB converters for underwater inductive charging. We employ LLC topology, offering superior soft-switching and theoretical CC–CV modeling.
- **Unnikrishna Menon et al. (2013)**: Proposed resonant magnetic induction for underground sensors. We tailor solutions to underwater AUVs, prioritizing converter integration and battery strategies.
- **Manikandan et al. (2018)**: Experimentally studied winding types and core shapes. Our simulation optimizes coil design and focuses on efficiency curves without physical prototypes.

Objectives

The objective of our project is to develop a Wireless Power Transfer (WPT) system along with Constant Current & Constant Voltage (CC-CV) Switching for Autonomous Underwater Vehicles (AUVs) to enable efficient and safe underwater charging. This system aims to:

- Reduce the dependency on manual assistance for battery replacement and charging operations.
- Enhance the operational autonomy and efficiency of AUVs by providing a faster initial charge upto 80% SoC, after which the system switches to Constant Voltage Mode in order to keep AUV Battery Health safe.
- Mitigate the risks associated with traditional charging methods, including connector and wire deterioration and the involvement of ship crews under challenging sea conditions.

Flowchart

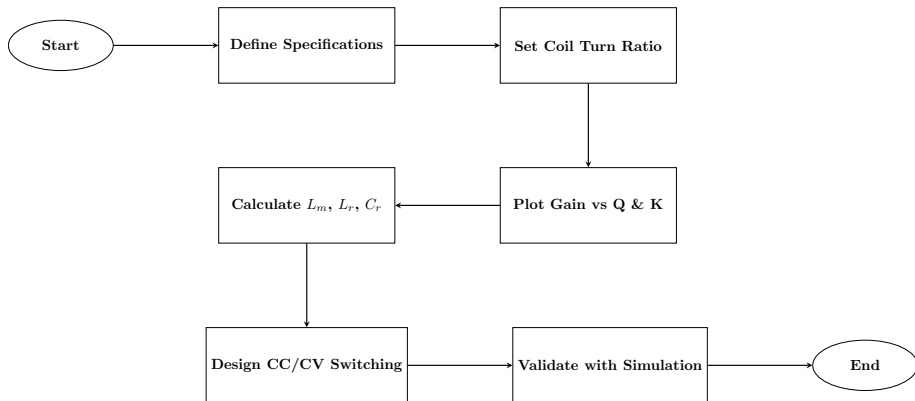


Figure: Flowchart

The wireless charging system consists of:

- 1 **Transmitter:** Full-bridge LLC resonant converter with submerged coupling coils, powered by a floating photovoltaic system and battery storage.
- 2 **Receiver:** Coupling coils connected to a rectifier and DC-DC buck converter to regulate output voltage for AUV compatibility.
- 3 **Battery Charging:** CC–CV control managing charging — constant current up to 80% SOC, then switching to constant voltage mode until full charge.

Component Selection:

- Resonant inductance:

$$L_r = \frac{V_{dc}}{4 \times f \times \Delta I_L} = 72 \mu\text{H}$$

- Magnetizing inductance:

$$L_m = K \times L_r = 360 \mu\text{H}$$

- Resonant capacitance:

$$C_r = \left(\frac{1}{2\pi f_0 \sqrt{L_r}} \right)^2 = 72 \text{ nF}$$

LLC Converter Charging Topology

The **LLC resonant converter** is a highly efficient and popular topology used in modern power conversion systems, especially for **battery charging applications**. Its ability to achieve **zero-voltage switching (ZVS)** significantly reduces switching losses, making it ideal for **high-frequency, high-power-density systems** like **electric vehicle (EV) chargers**, **robotic docking stations**, and **wireless charging platforms**.

Benefits of LLC Converters:

- **High Efficiency** ($>96\%$) even at varying loads
- **Zero Voltage Switching (ZVS)** for reduced switching losses
- **High Power Density** enabling compact charger designs
- **Lower EMI** due to soft switching
- **Wide Input and Load Range** capability
- **Scalable Design** for small devices to high-power EV chargers

Topology Comparison

Feature	LLC	DAB	CLLC
Efficiency	Very High (96–98%)	High (90–95%)	Very High (96–98%)
ZVS Capability	Excellent	Good	Excellent
Power Density	High	Medium-High	High
Control Complexity	Moderate	High	High
Operating Frequency	High (100–500kHz)	Medium (20–200kHz)	High (100Khz–200Khz)
Charging Suitability	EVs, Batteries, Robotics	EVs, Batteries	Wireless Charging

Table: Comparison of LLC Converter Topologies vs Other Charging Topologies

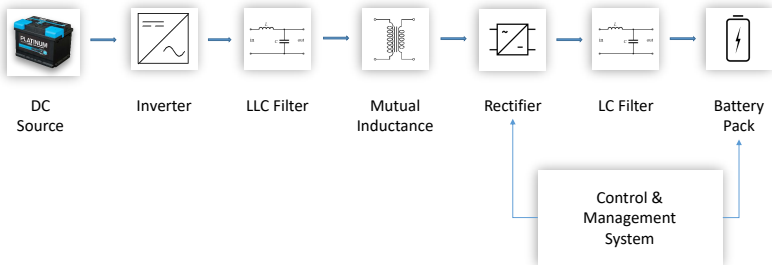
Introduction to CC–CV Charging

- CC–CV charging is a two-stage method where charging starts with a fixed current and then transitions to a fixed voltage.
- In **CC Mode**, the battery charges at constant current up to approximately 80% State of Charge (SOC), allowing the voltage to rise gradually.
- In **CV Mode**, once the battery reaches the set voltage limit, the voltage is held constant while the current gradually decreases until the battery reaches 100% SOC.
- This approach ensures fast, efficient charging while protecting the battery from overvoltage, overheating, and degradation.

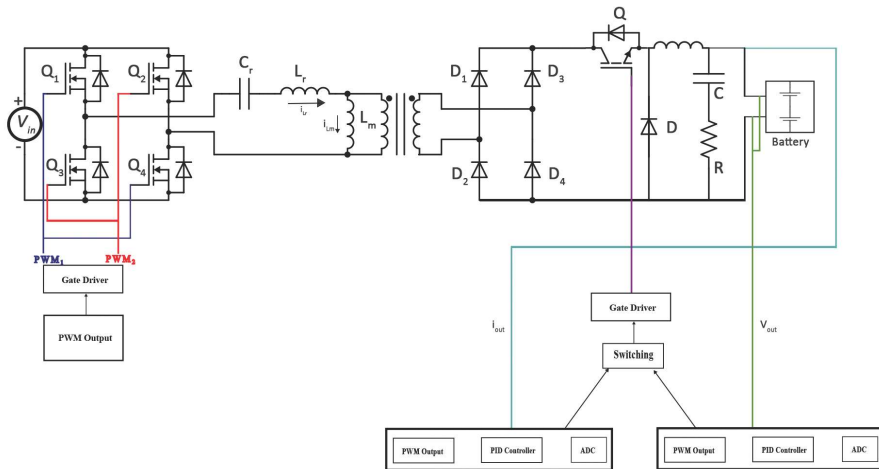
Benefits of Using CC–CV Charging

- **Improved Battery Life:** Reduces internal stresses and extends battery lifespan.
- **Optimized Efficiency:** Maximizes energy transfer with minimal losses.
- **Temperature Control:** Limits overheating risks during the CV phase.
- **Enhanced Safety:** Prevents overcurrent, overvoltage, and catastrophic failures.
- **Stable Performance:** Maintains battery operation within safe voltage-current limits for long-term underwater deployments.

Block Diagram



Circuit Diagram

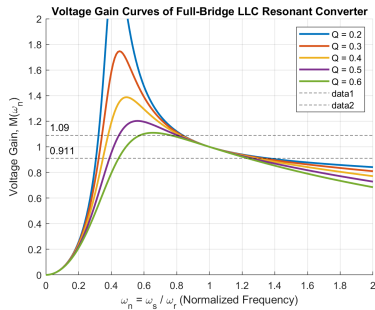


Voltage Gain

FORMULA

$$M = \frac{v_o}{v_{in}} = \frac{1}{\sqrt{\left\{ \frac{1}{K} \left[1 + K - \left(\frac{\omega_r}{\omega_s} \right)^2 \right]^2 \right\} + \left[Q \left(\frac{\omega_s}{\omega_r} - \frac{\omega_r}{\omega_s} \right) \right]^2}}$$

OUTPUT CURVE



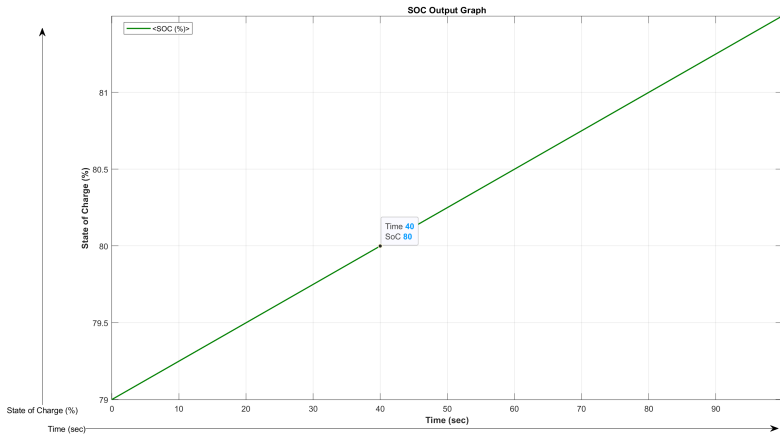
Implementation & Parameters

- 1 **System Design:** Simulated individual modules—LLC resonant converter, rectifier, buck converter & CC-CV switching Module.
- 2 **Parameter Tuning:** Adjusted resonant frequency (100 kHz) and coupling coefficients.
- 3 **Simulation Setup:** Modeled the water medium (air, freshwater, seawater) to assess performance variations.

Specifications	
Max. Input Voltage (V_{inmax})	170 V
Nominal Input Voltage (V_{in})	155 V
Min. Input Voltage (V_{inmin})	141 V
Output Voltage	48 V
Rated Power	200 W
Resonant (f_r) & Switching Frequency (f_s)	100 KHz
Resonant Inductance	72 μ H
Magnetizing Inductance	360 μ H
Resonant Capacitance	72 nF

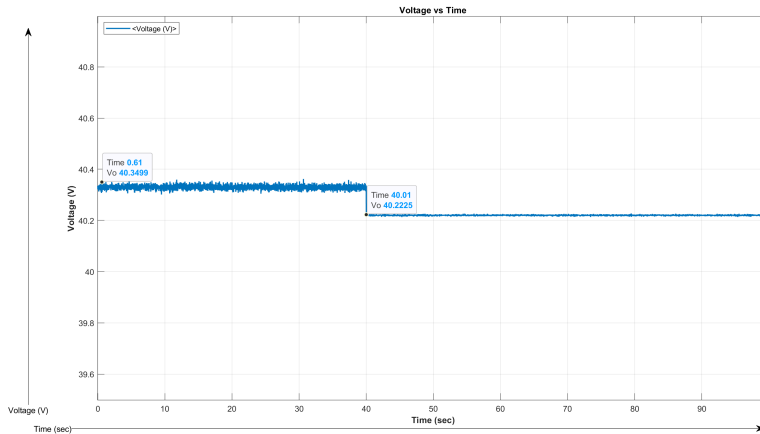
Final Results

State of Charge Graph



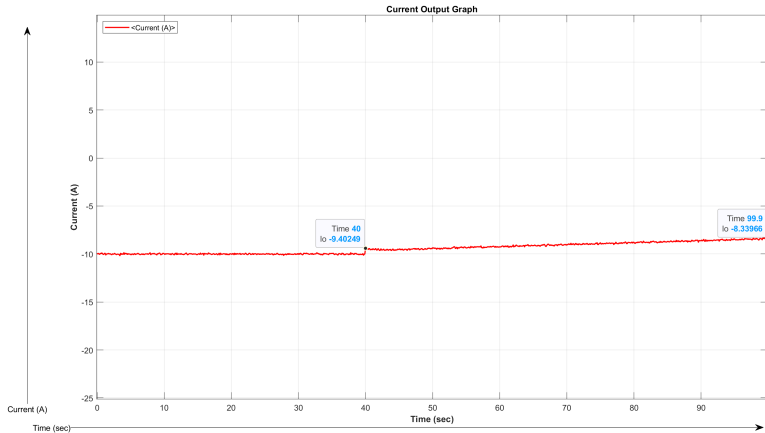
Final Results

Output Voltage Graph



Final Results

Output Current Graph



Conclusion

Our research presents a significant advancement in underwater exploration by addressing the primary limitation of AUV operations—inefficient battery charging. Through the design and implementation of a wireless power transfer (WPT) system featuring a full-bridge LLC resonant converter and CC-CV charging strategies, we ensure high energy transfer efficiency and safe battery management. This work reduces operational downtime, extends AUV endurance, and lays the groundwork for future intelligent underwater power systems, with promising applications in deep-sea exploration and environmental studies.

- **Hardware Prototyping:** Build a physical prototype to validate simulation results and address practical issues such as EMI shielding, coil alignment, and thermal effects.
- **Real-World Environmental Modeling:** Factor in variations like temperature, pressure, salinity, and wave-induced coil misalignments for improved system simulations.
- **Dynamic Impedance Matching:** Develop real-time algorithms to optimize system efficiency in fluctuating underwater conditions.
- **Multi-AUV Charging:** Explore frameworks for sequential or parallel charging with intelligent scheduling to support large-scale deployments.
- **Battery Optimization:** Integrate battery degradation models to enhance charging algorithms and extend battery service life.

The End